

1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Experiment 1.1

Generation of Continuous Time Signals

Program:

```
% Generation of Continuous Time Signals
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = -5:0.01:5;
```

```
% Continuous-time sine signal
```

```
x = sin(2*pi*t);
```

```
% Plotting
```

```
figure;
```

```
plot(t, x, 'LineWidth', 2);
```

```
grid on;
```

```
title('Continuous Time Sine Signal');
```

```
xlabel('Time (seconds)');
```

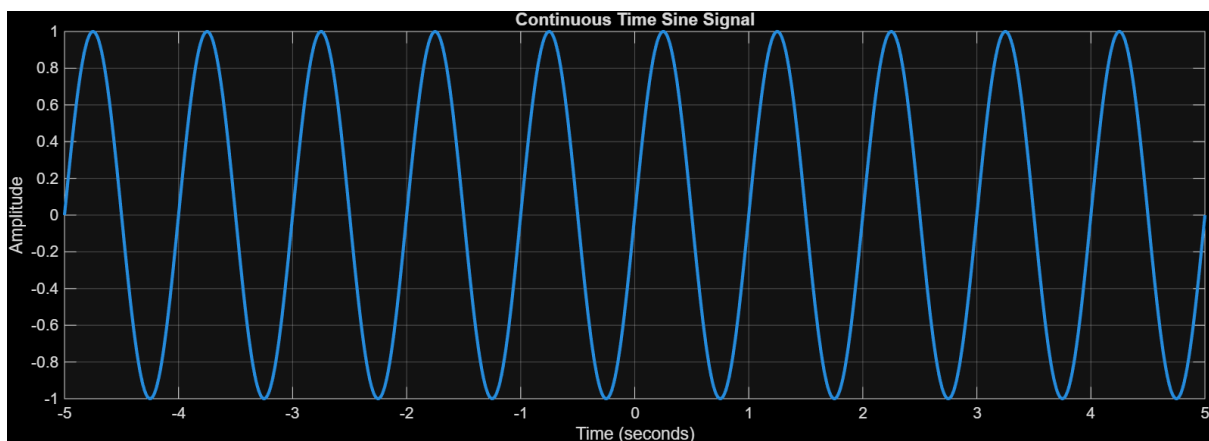
```
ylabel('Amplitude');
```

Expected Output:

A continuous-time sine wave is displayed with:

- Time range: -5 s to 5 s
- Amplitude range: -1 to 1
- Periodic sinusoidal waveform

Output Waveform:



Result:

This test case validates that the signal generator can produce a sinusoidal waveform with

Experiment 1.2

Generation of Discrete Time Signals

Program:

```
% Generation of Discrete Time Signal
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Discrete-time index
```

```
n = -10:10;
```

```
% Discrete-time sinusoidal signal
```

```
x = sin(0.2*pi*n);
```

```
% Plotting
```

```
figure;
```

```
stem(n, x, 'filled');
```

```
grid on;
```

```
title('Discrete Time Sinusoidal Signal');
```

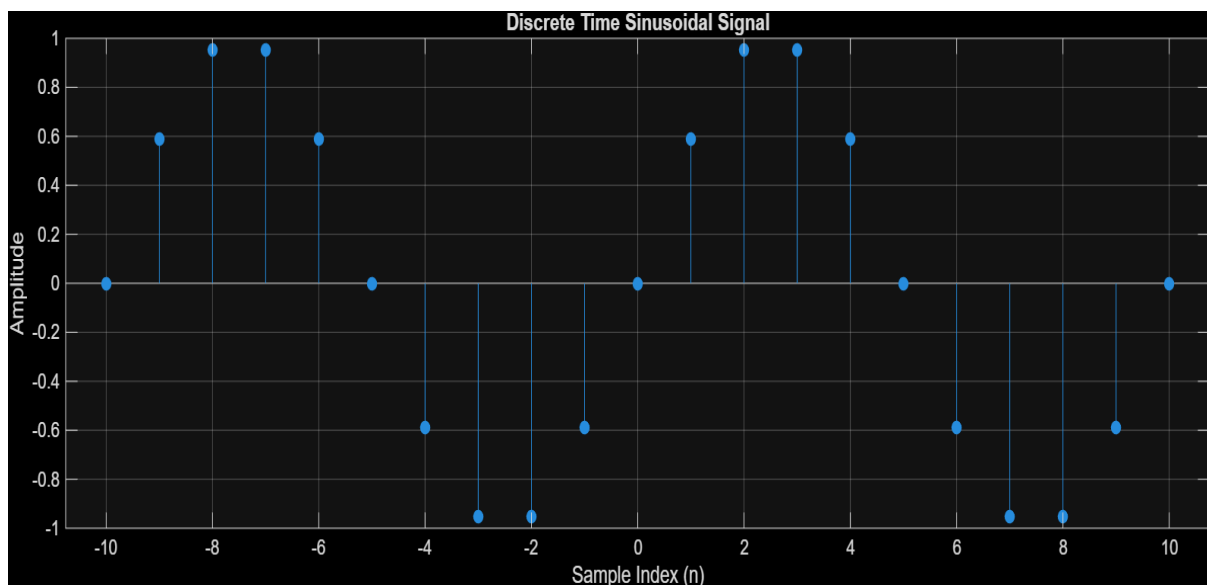
```
xlabel('Sample Index (n)');
```

```
ylabel('Amplitude');
```

Expected Output:

A discrete-time sinusoidal signal is displayed using stem representation, showing individual samples at discrete time instants.

Output Waveform:



Result:

This test case validates that the signal generator can accurately produce a discrete-time unit impulse signal at a specified index, which is essential for analyzing the response of discrete time systems.